



Figure 1: UML conceptual diagram

```
surya@DELL-Rnd-India:/code/kernel/lfy/linux-2.6.35-rc3$ pwd
/code/kernel/lfy/linux-2.6.35-rc3
surya@DELL-Rnd-India:/code/kernel/lfy/linux-2.6.35-rc3$ ARCH=um make menuconfig
```

Figure 2: Kernel configuration for UM architecture

```
File Edit View Terminal Help
.config - Linux Kernel v2.6.35-rc3 Configuration

Linux Kernel Configuration
Arrow keys navigate the menu. <Enter> selects submenus -->. Highlighted letters are
hotkeys. Pressing <h> includes, <u> excludes, <u> modularizes features. Press
<Esc> to exit, <?> for help, </> for Search. Legend: [*] built-in [ ] excluded
<u> module <u> module capable

[*] UML-specific options ---
[*] <u> enable loadable module support --->
[*] <u> enable the block layer --->
[*] <u> block devices --->
    character devices --->
    generic driver options --->
[*] <u> networking support --->
    file systems --->
    security options --->
    cryptographic API --->
    library routines --->
    SCSI device support --->
    [*] Multiple devices driver support (RAID and LVM) --->
v(y)

<Select> <Exit> <Help>
```

Figure 3: The 'make menuconfig' ncurses-based interface

```
File Edit View Terminal Help
.config - Linux Kernel v2.6.35-rc3 Configuration

Linux Kernel Configuration
Arrow keys navigate the menu. <Enter> selects submenus -->. Highlighted
letters are hotkeys. Pressing <h> includes, <u> excludes, <u> modularizes
features. Press <Esc> to exit, <?> for help, </> for Search. Legend: [*]
built-in [ ] excluded <u> module <u> module capable

[*] <u> subject debugging
[*] <u> Compile the kernel with debug info
v(y)

<Select> <Exit> <Help>
```

Figure 4: Selection of debug options

```
surya@DELL-Rnd-India:/code/kernel/lfy/linux-2.6.35-rc3$ ls
arch      Documentation  init      linux      modules.order  samples    tools
block     drivers        ipc       MAINTAINERS  Module.symvers  scripts    usr
COPYING   firmware      Kbuild   Makefile    net             security   virt
CREDITS   fs             kernel   mm           README          sound      vmlinux
crypto    include       lib       modules.builtin  REPORTING-BUGS  System.map vmlinux.o

surya@DELL-Rnd-India:/code/kernel/lfy/linux-2.6.35-rc3$ file linux
linux: ELF 64-bit LSB executable, x86-64, version 1 (SYSV), dynamically linked (uses shared libs), for
GNU/Linux 2.6.15, not stripped
surya@DELL-Rnd-India:/code/kernel/lfy/linux-2.6.35-rc3$
```

Figure 5: List of files after kernel compilation

- A root file system (can be created, or you can download one from http://uml.nagafix.co.uk/FedoraCore6/FedoraCore6-AMD64-root_fs.bz2. Creating a rootfs from scratch is beyond the scope of this article.)

Building the UML kernel and modules

Extract the kernel source archive:

```
$ tar -jxvf linux-2.6.35-rc3.tar.bz2
```

Enter the directory in which it was extracted (I'm using `/code/kernel/lfy/`), and issue the following command to compile the kernel for the UML architecture (view Figure 2).

```
$ ARCH=um make menuconfig
```

`ARCH` defines the architecture for which the kernel is to be compiled; in this case, 'um' stands for user mode.

The `make menuconfig` command gives us a `ncurses`-based interface in which we can configure build options for the UML kernel. See Figure 3.

To enable us to debug the kernel, we need to enable the following options:

- Compile the kernel with debugging info (see Figure 4).
- Compile the kernel with frame pointers.
- 'Enable loadable module support' in the main menu.

Once you've chosen these options and saved the configuration, proceed with kernel compilation. (If you aren't familiar with the process in general, you may want to refer to one of the many "kernel compilation how-to" pages on the Internet.) Issue this specific `make` command to compile the kernel:

```
$ ARCH=um make
```

Once the kernel compilation is done, you should have a binary named `linux` created in the same directory; see Figure 5.

Kernel modules need to be installed in a directory (of the host system), so that we can later copy them to the `/lib/modules/` path inside the UML system. In my case, the target directory is `/code/kernel/lfy/mods`.

```
$ ARCH=um make modules_install INSTALL_MOD_PATH=/code/kernel/lfy/mods
```